dMFM / multiMFM: status of every experiment

Reproducibility round, 2026-09-29/30 — manuscript revision marked up

Generated October 1, 2026

Every claim in the submission, its current verdict, and where the evidence lives. Verdicts are from first-hand measurement; where something is still running it says so rather than predicting. Repository: branch `reproducibility-package`, 28/28 tests passing. Detail: `docs/STATE_OF_CLAIMS.md`.

1 One-line summary

- **The DNA and QM9 results hold**, and five of them *improve* once dMFM is run at its own operating point. Table 2 was already optimal.
- **Two of the paper's most exposed problems were our errors**, not the method's, and are withdrawn: "an accurate posterior does not guide" and the dMFM–GLASS deficit.
- **Table 3 (GEOM) does not survive.** The negative is now explained mechanistically, and what replaces it is a sharper critique of the benchmark.
- **Four items need correcting regardless of framing** — they are factual, not defensive (§6).
- **The manuscript is now marked up and rendered** (§2): every table that was wrong is replaced, every figure comes from a rerun, and ten main-text changes are coloured — red to remove, blue to insert — rather than applied silently.

2 What changed in the manuscript today

`iclr2027_conferencemod.tex` renders from this repository alone: 38 pages, no undefined references. **GEOM is now the only item in the paper still carrying a marker.** Full list with sources: `docs/paper_changes_redblue.md`.

Item	Status	What was done
Table 1	↑ replaced	$1.174 \to 0.153$ and $0.833 \to 0.117$ at 1 step × MC 32, n=1000; 98.3% / 97.2% of pairs improved
Tables 14–15	↑ replaced	**every cell**, now finite-N *dMFM* against the same GLASS-2048 reference. The submitted cells were GLASS-on-GLASS while the caption said dMFM. dMFM beats the printed numbers in nearly every cell, and error *falls* from 50 to 400 bp
Table 16	replaced	Pearson rows dropped (provably Uniform$(-1, 1)$); total-variation distance with a real-vs-real floor
Table 18	replaced	cosine similarity instead of MAE, because the reference is numerically degenerate (corr(MAE, estimate norm) 0.95–0.995)
Table 6	↑ replaced	parent-disjoint Evo 2 run; the direction-specific claim **reverses** and its magnitude is 16× smaller — claim dropped
Table 7	no change	weights already read 1.0/1.0
Figures 4–8	replaced	all from reruns; Fig 5 finally plots what its caption claims
Table 5	↑ replaced	recomputed on the same 1×32 samples, so Tables 1 and 4–6 now describe *one* set of samples. The submitted deficit for the negative target does not reproduce: $0.9507\times \to 0.9910\times$ pairwise and $0.8027\times \to 0.9508\times$ nearest-generated, about five times less clustering

Two of the ten text edits are factual corrections, not framing. Eq (17) is not the reward that was optimised — the sampler exponentiates $-11.1(f - y^\star)^2$, an inverse temperature that changes the gradient *direction* and leaves ESS 1–3 at N=8. And the Fig 4 low-NFE claim was unsupported: the four baseline curves were hard-coded in the plotting script and never measured. The replacement states the measured result *including where the baselines overtake us* (Feynman–Kac is ahead by 824 NFE).

The paper is stand-alone. Nine passages compared it with an earlier version of itself ("the submitted table reported", "the originally published panels", "does not reproduce"). All are gone; the rendered PDF contains *zero* occurrences of *submitted*, *originally published*, *original submission*, *previous version* or *earlier version*. Facts those passages carried were kept and restated without the comparison. Comparisons to *other* papers (TFG-Flow's published values) are untouched.

The guidance strength is now stated as what it is. A new equation gives the coefficient the samplers actually use — one tuned scalar γ in place of $\sigma_t^2/2$ — and says plainly that we adopted it because it was *empirically more stable*, not because the theory asks for it, and that γ **forms no part of the theory**: the exact-law results are claimed only for Eqs (1), (2) and (16). The gap is quantified (0.009–0.58 at t=0.05 to 3–209 at t=0.95) so it cannot be mistaken for a reparameterisation, and Algorithms 2–3 now take γ as an input. **GEOM's six markers are the only ones left in the paper.**

Two files are local stand-ins, both marked as such in their own headers: `iclr2027_conference.sty` and `bibliography.bib` (placeholder entries carrying only the citation key; no bibliographic details are invented). Replace both from Overleaf before submitting.

3 Results that improve at the corrected operating point

The value gradient costs steps × MC network calls, so a budget can be split many ways. The split was chosen *once* on the Table 1 task and applied unchanged — not re-picked per table. The rule: *match the split to the split the student was distilled for.*

Item	As published	At the operating point	Significance
Table 1 / Fig 3	0.193 / 0.172	↑ **0.153 / 0.117**	**measured at** n=1000 [c]
Fig 7 (guide MAE)	0.158	↑ **0.047**	4.9 SEM [a]
Fig 7 (within 0.10)	0.53	↑ **0.86**	— [a]
Fig 8	0.172	↑ **0.076**	2.3σ
Table 13	+0.1262	↑ +0.1393	4.4 SEM (scale-free rule) [b]
Table 18 (`motif`)	0.1636 / cos 0.139	↑ **0.0479 / cos 0.318**	3.4× better MAE
Table 2 (QM9)	—	already optimal, no change	argmin by 2.2–6.2 SEM

[a] GPU, n=64 (Slurm 49465152); full-n confirmation is job 49476650. GLASS also prefers 2×16 to its own 4×8 ($-0.0294 \pm .0082$, 3.6 SEM). [c] Job 49461504, n=1000 per target, collected into `table1_dmfm_s1mc32_n1000_summary.json`; the 4×8 summary is kept separately. [b] GPU at the published n=100 (Slurm 49465154_0), which reproduces the published cell almost exactly: unguided 0.15229 vs 0.15229, guided 0.0261 [.0218, .0307] vs 0.0260 [.0215, .0306], paired +0.1262 vs +0.1263. At NFE 8 the published 1×8 beats 2×4/4×2/8×1 by 3.9/9.9/13.7 SEM; 8×1 under clip-down improves only **13 of 100** pairs, confirming that rule is inert there.

A coherence gain follows: every dMFM DNA result then uses the diagonal one-step student, so the paper describes one artifact rather than two.

An honest nuance on Fig 7. At each arm's *own* best split the two arms are a dead tie on the held-out oracle: dMFM 1×32 vs GLASS 2×16 is $+0.0225 \pm .0087$ on the guide (2.6 SEM, dMFM behind) but $+\mathbf{0.0012 \pm .0187}$ on the oracle (0.1 SEM). The correction removes an *artificial* 3.2× guide gap and leaves a real 1.9× one — and nothing at all on the independent evaluator.

4 Verdict per paper item

Item	Verdict	Evidence
Table 2, multiMFM	**holds**	No port bug; reproduces.
Table 2, steer-search	**holds**	Real 9–27% oracle gain, significant on all six paired (t = 4.5–36.9). μ, $\Delta\varepsilon$, LUMO are clear wins; Cv, α, HOMO are **ties** and should be reported as such. The 20× guide/oracle spread is winner's curse from best-of-3680, and the paper only ever published the oracle metric.
Table 1 / Fig 3 (dMFM)	**holds**	**Final, at the corrected operating point** (1 step × MC 32, n=1000 per target, job 49461504): $1.174 \to \mathbf{0.153}$ [.146, .161] and $0.833 \to \mathbf{0.117}$ [.111, .123], with **98.3%** and **97.2%** of pairs improved against a parent-disjoint oracle. Against matched GLASS-4 under the identical rule the paired difference is $+0.0007 \pm .0052$ and $\mathbf{-0.0064 \pm .0045}$ — indistinguishable at $y^\star$=+1 and nominally *ahead* at −1. Unguided arms bit-identical across arms, so the whole difference is the guidance term. For scale: the mis-wired configuration gave 0.610/0.408, and the published GLASS numbers were 0.148/0.119.
Fig 7, Fig 8	**holds**	Survive with dMFM; 12/14 ladder points in. Ladder is *not* monotone — best at gf 4, regressed by 32 — so a "strong guidance" panel at 32 is over-driven.
Table 4; Table 5 (+1); Table 6 ordering	**holds**	Survive.
Fig 4 left	**holds**	dMFM value error $4.80 \to 0.247$ over N=1 → 512, crossing the N-independent DPS baseline at $N \approx$ 8–16. Caveat: the reference's own self-split floor is 0.324, so $N \geq 256$ is floor-limited.
Fig 4 right	**holds** (narrow)	Holds at *one* operating point, not as a regime. On the measured forward-equivalent axis (k=1.1) the Feynman–Kac crossover is **430**, not 582, and the win region is MC=1 only. No wall-clock regime where dMFM wins.
Tables 13, 17, 18	**holds**	Used the dMFM students. Table 18's two mismatches are paper double-rounding.

Item	Verdict	Evidence
dMFM vs GLASS	**holds**	**Tie** at matched NFE once each arm uses its own split: dMFM 1×32 gives 0.146/0.120 vs GLASS's best 2×16 at 0.143/0.117 ($+0.004 \pm .006$, $+0.003 \pm .006$).
Table 5 (-1), Table 6 magnitudes	**open**	Do not reproduce, but our `split65k` attribution was **unsupported**: metric and Evo 2 code are AST-identical to the originals and the guidance-independent rows bracket the model move 24–66× below the effect. The difference is the guidance protocol (six knobs). Open pending job 49431930.
Table 16	**fails**	**Neither number is right.** The uniform-vs-real global r is provably Uniform$(-1, 1)$ — pure noise, checked against 20,000 draws. The printed 0.1813, the figure's -0.852 and our 0.4576 are three draws. Drop the rows or use total-variation distance.
Table 3 (GEOM)	**fails**	4096×3: gain -0.00031 ± 0.00065. Published figure is best-of-4-grid at n=64, one seed. Coordinate guidance fails across all 7 pre-registered regimes and 3.5 decades of step size.
Fig 1, Table 12 (toy)	**holds**	Rebuilt from transcript fragments and retrained; PDFs now embed TrueType (were Type 3, which arXiv/ICLR reject).

5 Withdrawn — these were our errors, not the method's

1. "A well-integrated posterior does not guide at all." Rested on one arm, `glass_euler32`, with two defects: it integrates to the *singular* data time t=1 (endpoint a simplex vertex, V piecewise constant, true fp32 gradient norm 1.2e−5 — the gradient was numerically *dead*), and it was the only arm never given the scale-free rule. Fixing both: published arm 0.8651 with 0/8 improved → **0.1939 with 8/8 improved**, against 0.1548 for the Table 1 estimator, on a 25× smaller gradient. The reference integrator delivers 89% of the coarse estimator's benefit.

2. The dMFM–GLASS deficit ($+0.041/+0.049$). Both arms were pinned to GLASS's 4×8 split — GLASS's estimator, not the flow map's. Since the value gradient costs steps × MC calls, a one-step flow map affords MC 32 at the same NFE. Parity is now confirmed *three* independent ways: Table 1 at n=600, Fig 7 on the held-out oracle, and Table 1 at n=1000 ($+0.0007 \pm .0052$, $-0.0064 \pm .0045$).

3. The QM9 DPS result ("DPS cannot use more compute"). The interpolant is a contraction, so delivered guidance scales as $1/N$: at 4096 steps it is 0.031× the 128-step dose. The degradation was a shrinking dose, not accurate integration. Dose-matched rerun queued (49455395).

All three share one root cause: guidance constants are not transportable across samplers or step counts without renormalisation. It has now produced three false findings and is the first thing to check when a guided arm looks dead.

6 Corrections that are factual, not framing

1. **Eq (17) is not the reward that was optimised.** The code exponentiates $-11.111(f - y^\star)^2$. Inside $\log \mathbb{E}[e^r]$ that constant is an inverse temperature: it changes the gradient *direction* and cannot be absorbed into the tuned strength. It drops the effective sample size of the N=8 draws to **1.09 / 1.98**, so "$N = 8$" overstates the averaging 3–7×; report "$N = 8$, ESS 1–3". The published strength cannot be re-derived from Eq (17) at all.
2. **"Our base generator beats TFG-Flow's guided" is unsound and must be dropped.** Random-pair FP2 Tanimoto on the val set is **0.2164**, so our 0.2278 and their 0.208 are *both* at the floor; and TFG-Flow averages over all samples, scoring invalid ones 0 and mapping MASK atoms to hydrogen.
3. **Eq (1) does not describe the DNA runs** — the implemented coefficient is 10–16× the stated $\sigma^2/2$.
4. **Table 16's rows are noise** (see above).

7 GEOM: what replaces Table 3

Three real defects were found, and the most promising rescue was tested and refuted.

- **Guidance is annihilated at t=1.** Both interpolants contract onto the endpoint prediction with coefficient exactly 1, so a perturbation injected anywhere in the window survives with magnitude **0** (coords) / $\leq$3e−11 (atom types). The earlier "0 argmax flips in 420 atoms" was a defect of *where* guidance is applied.
- **The reward is a poor surrogate for the metric**: Spearman 0.36–0.38 for $-\|p-c\|^2$ against 0.67–0.71 for a differentiable Tanimoto.
- **The composition rescue is refuted at n=256.** One head-guided substitution gains +0.1005 Tanimoto but PoseBusters collapses 0.879 → 0.152 — and *random* substitution destroys validity just as fast (0.367), so the cause is the substitution operator (wrong bond lengths at fixed geometry), not the head.
- **The salvageable contribution:** over half the attainable headroom (+0.100 of +0.18) can be taken by relabelling ∼15% of atoms while leaving 85% of molecules invalid. A structural-fingerprint Tanimoto is therefore trivially gameable by a method that never touches geometry, and **any number on this benchmark must be reported with a validity check** — which TFG-Flow's Table 3 does not do.

8 Everything reproduces from the shipped checkpoints

`notebooks/reproduce_from_checkpoints.ipynb` re-runs *inference* for nine paper items from `checkpoints/` and compares each recomputed number with the one it should reproduce. No training, and nothing is read back from a stored result and passed off as a rerun — stored values appear only in the *expected* column. Both executions passed end to end: the CPU subset on a login node (292 s) and the full set on one A100 (job 49533348, **23 min 32 s**). **157 comparisons: 130 match, 27 explained.**

The two numbers the paper leans on hardest reproduce from checkpoints on a fresh GPU:

- **Table 1's corrected cells.** At n=64, $y^\star$=−1 guided oracle MAE **0.1167** against the 0.117 headline; $y^\star$=+1 inside its CI; both unguided arms to 10^{-3}; 8/8 match.
- **Table 18's degenerate reference** — the evidence for reporting cosine instead of MAE. Measured corr(MAE, ‖estimate‖) is **0.967** on `gc` and **0.996** on `motif`, with reference norm 0.0056–0.069.

Table 13 (8/8), Table 16 (9/9 at n=4096), Table 11 (8/8 to the printed digit) and QM9 Table 2 (34/36) also pass. The 27 non-matches are all accounted for: 16 compare parent-disjoint results against the purged `split65k` paper values, 9 are small subsamples judged against a full-grid mean, and 2 are QM9 SS rows against the published base-endpoint harvest. **None is a failed reproduction.**

9 Closed: the student is not undertrained

The most obvious alternative explanation for the DNA numbers — that the shipped student (`epoch=29-step=15000`) was simply selected at the wrong point — was tested and ruled out. Retrained for **60,000** steps with checkpoints kept by *validation loss* rather than step count (Slurm 49471652), then evaluated against the shipped artifact on the *same* 200 paired samples at the same operating point (Slurm 49519367): paired oracle MAE, retrained − shipped, is $+\mathbf{0.0020} \pm \mathbf{0.0133}$ at $y^\star$=+1 and $+\mathbf{0.0009} \pm \mathbf{0.0114}$ at −1 — 0.15 and 0.07 SEM. Four times the training and a principled selection rule move nothing, so the shipped checkpoint stays and every DNA result continues to come from one artifact. Detail: `results/dna/deficit_rescue/student_retrained_L50_n200/VERDICT.m`

10 In flight

Job	What	Why it matters
49455395	Dose-matched DPS	7 of 10 tasks completed; decides whether the withdrawn DPS effect exists

11 Decisions for the authors

1. **Table 3:** reframe as a negative result plus its mechanism (coordinate guidance cannot steer graph topology; the benchmark is gameable without a validity check), or drop it.
2. **Table 18's metric.** Its reference has gradient norm 0.006–0.090 against estimates of 0.18–7.2, with $\mathrm{corr}(\mathrm{MAE}, \text{estimate norm}) = 0.95$ on `gc` and **0.995** on `motif`. The table largely measures gradient *magnitude*, not accuracy.
3. **Table 2 SS row:** report three wins and three ties, not six wins.
4. **Fig 4's axis:** ship the measured forward-equivalent axis (crossover 430) or the published one (582)? Both versions exist.
5. **Co-author review** of all of the above.
6. **The red/blue markup**: accept, reject or amend each of the fourteen marked changes (`docs/paper_changes_redblue.m`). Two are factual corrections rather than framing and should not simply be rejected.